

A02 - Deterministic Fault Campaign

Deterministic fault-injection campaign covering Host timeout, controller-side NAND failures, data-integrity mismatch detection, firmware-log correlation, reset, retry, and readback validation.

PASS

9	9	9	0
TOTAL	EXECUTED	PASSED	FAILED
0	0	0	7.387 s
ERROR	ABORTED	NOT RUN	DURATION

Suite ID	A02
Suite Name	Automated Fault Injection Campaign
Generated	2026-09-22 01:10:48
Fault Profiles	4
Storage Reset at Start	true
Storage Reset After Campaign	false

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Campaign Matrix

Case	Fault	Cmd	LBA	Expected Behavior	Recovery	Result
A02-FP01-C01	timeout	READ	0	Host TIMEOUT -> Reset -> Retry READ	Reset + Retry	PASS
A02-FP01-C02	timeout	READ	100	Host TIMEOUT -> Reset -> Retry READ	Reset + Retry	PASS
A02-FP01-C03	timeout	WRITE	200	Host TIMEOUT -> Reset -> Retry WRITE -> Readback	Reset + Retry + Readback	PASS
A02-FP02-C01	NAND_READ_FAIL	READ	300	FAILED CQ + NAND_READ_FAIL + FW log correlation	Log correlation / No retry	PASS
A02-FP02-C02	NAND_READ_FAIL	READ	1500	FAILED CQ + NAND_READ_FAIL + FW log correlation	Log correlation / No retry	PASS
A02-FP03-C01	NAND_PROGRAM_FAIL	WRITE	400	FAILED CQ + storage unchanged + FW log correlation -> Reset -> Retry WRITE -> Readback	Correlate + Reset + Retry + Readback	PASS
A02-FP03-C02	NAND_PROGRAM_FAIL	WRITE	2000	FAILED CQ + storage unchanged + FW log correlation -> Reset -> Retry WRITE -> Readback	Correlate + Reset + Retry + Readback	PASS
A02-FP04-C01	miscompare	READ	500	CQ SUCCESS + data mismatch detected; storage unchanged	None	PASS
A02-FP04-C02	miscompare	READ	2500	CQ SUCCESS + data mismatch detected; storage unchanged	None	PASS

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Fault Coverage Summary

Profile	Fault Type	Total	Exec	Pass	Fail	Err	Abort	Not Run	Validation Focus
FP01	Command Timeout	3	3	3	0	0	0	0	Host-side timeout detection followed by Controller Reset and retry with preserved CID progression.
FP02	NAND Read Failure	2	2	2	0	0	0	0	Controller returns FAILED / NAND_READ_FAIL and runtime firmware log is correlated to the command.
FP03	NAND Program Failure	2	2	2	0	0	0	0	Controller returns FAILED / NAND_PROGRAM_FAIL, failed write does not modify storage, then reset / retry / readback recovers.
FP04	Data Miscompare	2	2	2	0	0	0	0	CQ remains SUCCESS while returned data is corrupted; the validator detects the mismatch and storage remains intact.

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Campaign Case Evidence

FP01 - Command Timeout

Host-side timeout detection followed by Controller Reset and retry with preserved CID progression.

A02-FP01-C01 READ LBA 0	PASS
Fault	timeout
Expected Detection	TIMEOUT
Recovery	Reset + Retry
Validation Checks	16 / 16 PASS
Duration	2.129 s
Fault Result	CID 1 / TIMEOUT / Outstanding [1]
Queue After Timeout	SQ 1/1, CQ 0/0
After Reset	Outstanding [] / next CID 2
Retry READ	CID 2 / SUCCESS / pattern=None / unwritten=True

A02-FP01-C02 READ LBA 100	PASS
Fault	timeout
Expected Detection	TIMEOUT
Recovery	Reset + Retry
Validation Checks	16 / 16 PASS
Duration	2.132 s
Fault Result	CID 1 / TIMEOUT / Outstanding [1]
Queue After Timeout	SQ 1/1, CQ 0/0
After Reset	Outstanding [] / next CID 2
Retry READ	CID 2 / SUCCESS / pattern=TEST_PATTERN_A / unwritten=False

A02-FP01-C03 WRITE LBA 200	PASS
Fault	timeout
Expected Detection	TIMEOUT
Recovery	Reset + Retry + Readback
Validation Checks	18 / 18 PASS
Duration	2.185 s
Fault Result	CID 1 / TIMEOUT / Outstanding [1]
Queue After Timeout	SQ 1/1, CQ 0/0
After Reset	Outstanding [] / next CID 2
Retry WRITE	CID 2 / SUCCESS
Recovery Readback	CID 3 / NEW_PATTERN

FP02 - NAND Read Failure

Controller returns FAILED / NAND_READ_FAIL and runtime firmware log is correlated to the command.

A02-FP02-C01 READ LBA 300	PASS
Fault	NAND_READ_FAIL
Expected Detection	FAILED_CQ_AND_LOG
Recovery	Log correlation / No retry
Validation Checks	14 / 14 PASS
Duration	0.105 s
CQ Completion	FAILED / NAND_READ_FAIL
Firmware Log	CID 1 / READ / LBA 300 / NAND_READ_FAIL
Correlation	CID + Opcode + LBA + Status + Error matched

A02-FP02-C02 READ LBA 1500	PASS
Fault	NAND_READ_FAIL
Expected Detection	FAILED_CQ_AND_LOG
Recovery	Log correlation / No retry
Validation Checks	14 / 14 PASS
Duration	0.106 s
CQ Completion	FAILED / NAND_READ_FAIL
Firmware Log	CID 1 / READ / LBA 1500 / NAND_READ_FAIL
Correlation	CID + Opcode + LBA + Status + Error matched

FP03 - NAND Program Failure

Controller returns FAILED / NAND_PROGRAM_FAIL, failed write does not modify storage, then reset / retry / readback recovers.

A02-FP03-C01 WRITE LBA 400	PASS
Fault	NAND_PROGRAM_FAIL
Expected Detection	FAILED_CQ_AND_LOG
Recovery	Correlate + Reset + Retry + Readback
Validation Checks	22 / 22 PASS
Duration	0.209 s
Fault Completion	FAILED / NAND_PROGRAM_FAIL
Firmware Log	CID 1 / WRITE / LBA 400 / NAND_PROGRAM_FAIL
After Reset	next CID 2
Retry WRITE	CID 2 / SUCCESS
Readback	CID 3 / NEW_PATTERN

A02-FP03-C02 WRITE LBA 2000	PASS
Fault	NAND_PROGRAM_FAIL
Expected Detection	FAILED_CQ_AND_LOG
Recovery	Correlate + Reset + Retry + Readback
Validation Checks	22 / 22 PASS
Duration	0.209 s
Fault Completion	FAILED / NAND_PROGRAM_FAIL
Firmware Log	CID 1 / WRITE / LBA 2000 / NAND_PROGRAM_FAIL
After Reset	next CID 2
Retry WRITE	CID 2 / SUCCESS
Readback	CID 3 / NEW_PATTERN

FP04 - Data Miscompare

CQ remains SUCCESS while returned data is corrupted; the validator detects the mismatch and storage remains intact.

A02-FP04-C01 READ LBA 500	PASS
Fault	miscompare
Expected Detection	DATA_MISCOMPARE
Recovery	None
Validation Checks	9 / 9 PASS
Duration	0.156 s
Fault CQ Status	SUCCESS
Returned Pattern	TEST_PATTERN_B
Stored Pattern	TEST_PATTERN_A
Interpretation	CQ completed successfully; data validator detected returned-data mismatch.

A02-FP04-C02 READ LBA 2500	PASS
Fault	miscompare
Expected Detection	DATA_MISCOMPARE
Recovery	None
Validation Checks	9 / 9 PASS
Duration	0.156 s
Fault CQ Status	SUCCESS
Returned Pattern	TEST_PATTERN_B
Stored Pattern	TEST_PATTERN_A
Interpretation	CQ completed successfully; data validator detected returned-data mismatch.

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Campaign Conclusions

Host Detection	Timeout faults are detected by the Host without a normal CQ completion.
Controller Error Detection	NAND_READ_FAIL and NAND_PROGRAM_FAIL surface as deterministic FAILED completions.
Firmware Log Correlation	Controller-side NAND failures are correlated by CID, opcode, LBA, status, and error.
Data Integrity Validation	Miscompare cases preserve CQ SUCCESS while the validator detects returned-data mismatch.
Recovery	Controller Reset clears queue/outstanding state, preserves CID progression, and recovery uses a new CID.
Data Protection	Failed or timed-out WRITE paths do not modify storage before successful retry.
Campaign Result	All configured deterministic fault cases passed.

STOP v1: Cancellation is cooperative. Active fake_nvme subprocesses can be terminated and recovered through simulated controller reset; arbitrary Python deadlocks or non-cooperative infinite loops are outside the v1 stop guarantee.

Scope: This is a deterministic Host-side logical NVMe validation campaign. It demonstrates automation, fault detection, recovery reasoning, firmware-log correlation, and data-integrity validation. It does not claim physical SSD firmware or PCIe hardware validation.